

The evolution and analysis of action switching in embodied agents

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Abstract

The dynamical systems approach in the cognitive and behavioral sciences studies how systems made of many coupled components across brain, body, and environment self-organize to generate behavior. This approach has mostly focused on models of single actions and has not addressed how a dynamical system can engage in multiple different directed actions. In this paper, we introduce a family of artificial life models that demonstrate how dynamical agents can engage in multiple different actions and autonomously switch between them. These described agents engage in a food foraging task, and are driven by both internal, metabolic variables and external, sensory variables. The analysis of one of these agents demonstrates how different actions can arise through transient modes of sensorimotor coordination, in which a subset of the available sensors and effectors become engaged while others are ignored. Transitions between actions are analyzed and shown to correspond to rapid movements through the agent's state space. In these transitions, some of the previously controlling sensors and effectors disengage, and new sets of sensors and effectors are engaged.

Keywords

Action selection, embodied dynamicism, dynamical systems, sensorimotor invariants

1 Introduction

All organisms rely on the ability to engage in a broad repertoire of actions, and to flexibly switch between them in context-appropriate manners. For example, a subtle movement in a distant shrub might be all the information available for a small monkey, who is presently gathering berries off the ground, to determine whether a predator is on the prowl. This movement in the environment stimulates a monkey's relevant sensors in a way that elicits a dramatic behavioral response, transitioning the whole organism's sensorimotor system from gathering food to scampering up a tree for safety. Actions such as these require similar sets of body parts and neuronal groups, but engage them in different patterns of coordination. Picking food off the ground requires fine finger dexterity and acute eye movements for examining food sources, while climbing up a tree requires full limb coordination along with tactile and proprioceptive sensory input.

The problem of how agents are able to enact one action rather than another, and switch between them appropriately is usually addressed in the field known as action selection (Seth, 2007). This is a rich field whose history goes back to ethology, which originated in the 1930s. Ethological approaches (Lorenz, 1957; Tinbergen, 1950) often assume a hierarchical process in

which actions are defined as behavioral units, which are selected between by a higher-level decision-making mechanism, and once selected they unfold in "fixed action patterns." In much of the literature on decision-making, possible actions are evaluated given the available information from internal and external sources, and after a process of arbitration, the action with the highest probable beneficial value is selected and executed (Schall, 2001).

Later research, influenced by the paradigm of parallel distributed computing, refined the notion of hierarchical control, incorporated more parallelized architectures (Bryson, 2000; Tyrrell, 1993), and demonstrated the selection of action through spreading activation across internal representations of behavior (Maes, 1991). With the framework of embodied dynamicism, which stresses the importance of an agent's body and environment, it has been demonstrated that action selection does not require internal arbitration or representations of behavior, and can arise purely from

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sensorimotor interactions and the influence of an internal state (Seth, 1998).

In this paper, we adopt the perspective of embodied dynamicism, which sheds the assumptions of higher-level decision-making processes and instead commits only to dynamical systems explanations. This allows for more integrated architectures, which embrace nonlinearity and aim to describe behavior as emerging from rich interactions across the brain, body, and environment (Beer, 1995; Clark, 1997; Kelso, 1995; Port & Van Gelder, 1995; Van Gelder, 1998; Varela, Thompson, & Rosch, 1991).

These ideas are often demonstrated via computational models in the field of artificial life (Langton, 1989). The methodology of evolutionary robotics (Beer & Gallagher, 1992; Cliff, Husbands, & Harvey, 1993; Di Paolo, Noble, & Bullock, 2000; Harvey et al., 2005; Nolfi, Floreano, Miglino, & Mondada, 1994) has developed tools to evolve models of embodied dynamical agents that generate specified behaviors of interest. The resulting models are then analyzed with mathematical tools, such as dynamical systems theory, to uncover the underlying properties that allow for the self-organized behavior. Such analyses almost always discover strategies that human engineers would never have dreamt of, and demonstrate that cognition and behavior do not arise out of linear computational sequences, but instead embrace nonlinear processes and feedback between brain, body, and environment.

This paper aims to use this methodology to address what actions mean in the dynamical perspective, and how agents can autonomously transition between them. In this introduction, we begin with a common sense definition of action as a pattern of coordinated behavior instantiated by an agent, which robustly brings the system to some end-state, or goal. This is not quite the same as the fixed action pattern of ethology, because it is not necessarily a sequence of behaviors, but rather a mode of coordination. This qualitative definition is only preliminary, and through the analyses performed in this paper, we will build up a more explicit definition of action in terms of specific dynamical tendencies.

Most of the models generated by evolutionary robotics have examined instantiations of single actions. These include insect walking (Beer & Gallagher, 1992), visually guided movement (Cliff et al., 1993; Nolfi et al., 1994), and also actions that capture more complex cognitive process such as learning and memory (Izquierdo, Harvey, & Beer, 2008), selective attention (Slocum, Downey, & Beer, 2000), and relational categorization (Williams, Beer, & Gasser, 2008). There have also been several models that have begun investigating how dynamical agents might instantiate multiple different actions (Buckley, Fine, Bullock, & Paolo, 2008; Iizuka & Di Paolo, 2007; Izquierdo & Buhrmann, 2008; Seth, 1998).

The problem of action switching brings up new and interesting questions for the embodied dynamical

perspective. Different actions appear to be controlled in completely different ways, yet the perspective claims that they all arise out of the same organization of an agent with internal states, which is coupled to an environment. For the embodied dynamical perspective to be valid, the agent's behavioral repertoire and the transitions between them must arise from a system with a fixed number of variables and a single evolution operator that transforms their state with every time step. We must ask questions such as, "How does a system's organization bring forth different modes of coordination between the same, or different sets of body parts?" "How do radically different actions arise in otherwise similar conditions?" "How could a nervous system, which mediates between an organism's sensory inputs and motor outputs, constrain the sensorimotor apparatus in a way that generates unique modes of coordination?" "When an action is complete, how does the system transition to a different action?"

In Section 2, we introduce the methodology for evolving action-switching agents. In addition to sensors that can detect signals from the environment, we incorporate internal nutrient sensors that are sensitive to the internal metabolic states of the agent. The addition of metabolism-based sensitivities provides the agent with inputs that inform it about its internal state, allows it to direct its actions towards appropriate objects in the environment, and to adaptively regulate its internal state and sustain homeostasis.

In Section 3, we examine the strategies evolved by multiple different morphologies. Different morphologies lead to different dynamical strategies, and to be systematic about our investigation of dynamical strategies for action switching, this section provides a high-level view of action switching in the space of strategies that result from different morphologies. By examining multiple families of action-switching agents, we gain broader insight and a context for the next section's fine-grained analysis of a particular agent.

In Section 4, we select a particular action-switching agent, and analyze it in great depth to uncover its dynamical strategy. By analyzing this evolved agent with dynamical systems theory, we introduce a dynamical terminology into the study of multiple actions and switching. The results demonstrate how dynamical processes can lead to different modes of coordination (corresponding to different actions), and how the organization allows for autonomous switching between the actions.

Finally, in Section 5, we look at these results in order to interpret the embodied dynamical perspective on action switching, and how it relates to previous theories of action selection. The terminology of dynamical systems helps define actions more explicitly, and explains how their transitions arise in a single, nonlinear brain-body-environment system. Steps towards a more comprehensive understanding of action and switching are discussed.

2 Methods

The model of action switching used in this paper is based on a previous model of chemotaxis (Beer, 1995; Beer & Gallagher, 1992). In this original model, agents perform chemotactic behavior by moving up a chemical gradient towards the source of the chemical diffusion where the signal is most concentrated. These chemical sources will be referred to as resources. As an extension to the previous model, the agents presented in this paper are designed to approach two different resources that emit different chemical signals. These signals engage different sets of chemosensors, and therefore require different patterns of coordinated behavior as directed towards the different signals. The agents' task is to perform chemotaxis towards both resources, to gather nutrients from within the resources' boundaries, and to autonomously switch between the chemotactic actions. If agents are successful, we assume that they are able to engage in multiple actions due to the different modes of coordination required.

2.1 Morphologies and environment

To motivate the switch between approaches of the different resources, agents are given two nutrient levels that have to be sustained at a value above zero by autonomously switching between the approaches and retrievals of two different resources. The reason for including nutrient levels is to provide the agents with internal, motivational states that can be used to make distinctions about their condition. This is based in the same insight verified in a recent paper, which demonstrated that integrating internal, metabolism-based and external, environmentally-based sensitivities can lead to superior behavior (Egbert, 2013).

The agent's environment, shown in Figure 1, is a bounded 100×100 unit plane, with two resources that are held in fixed locations throughout each trial. These resources have a fixed radius of 7 units, within which the concentration of the chemical is fixed at a constant level. Outside of the resources' boundaries, the chemical signals diffuse through the environment with a smooth exponential decay from the center of the resource. More technical details are included in the appendix.

In this paper, we experiment with three different morphologies, shown in Figure 2. The morphologies differ in their number of chemosensors, the ways in which these sensors are spatially extended from the body, by what we will call chemosensor stalks, and whether or not there are nutrient sensors. Chemosensory input is based on the concentration of its particular type of chemical stimulant at the chemosensor location, and nutrient sensor inputs corresponds to the built-in nutrient levels.

The outputs of the two effectors produce forces that generate movement through the environment. Each output produces a vector of force that pushes

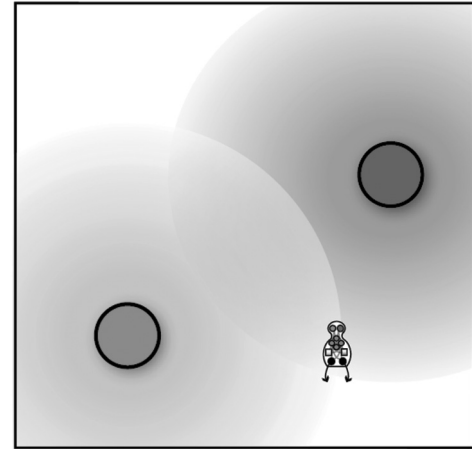


Figure 1. The agent's environment is a bounded two-dimensional space that contains two resources and the chemical gradients that surround them. The chemical concentration is flat within the resources' boundaries, and there is a smooth exponential decay of chemical concentration as a function of distance from the resource.

one side of the body, leading to a behavioral disposition that if the left motor neuron is more active than the right motor neuron, it turns the agent to the right (and vice versa). The two force vectors produced by the motor neurons' outputs are combined to determine overall direction and magnitude: directional change is determined by the difference between the two effectors' outputs, while magnitude is determined by the sum of the effectors' outputs. Because of this, when both effectors are equally active it results in the agent moving straight ahead. When one effector is more active than the other, it drives the agent along an arc, or into a spin if one effector is completely inactive. Additionally there is a frictional effect, so that agents slow down and eventually stop if the effectors produce no force. These laws of motion are described more extensively in the appendix.

The two nutrient levels take a continuous range of values between 0 and 10. A nutrient level is increased at a fixed eating rate when the agent is within the boundary of the nutrient's corresponding resource, and is drained by a constant metabolism no matter where the agents are positioned. The rates were designed to make it necessary for the agent to stop on top of each resource for an extended period before having to leave for the other resource in order to maintain the nutrient at a sufficient level.

The simulation continues as long as both nutrient values are above 0, but as soon as either value drops to 0 the agent dies and the simulation ends. If the agent is successful and does not die, the simulation is halted after 5000 time steps. If it survives for this long, it must have approached and eaten each resource multiple times and is therefore assumed to have successfully solved the task.

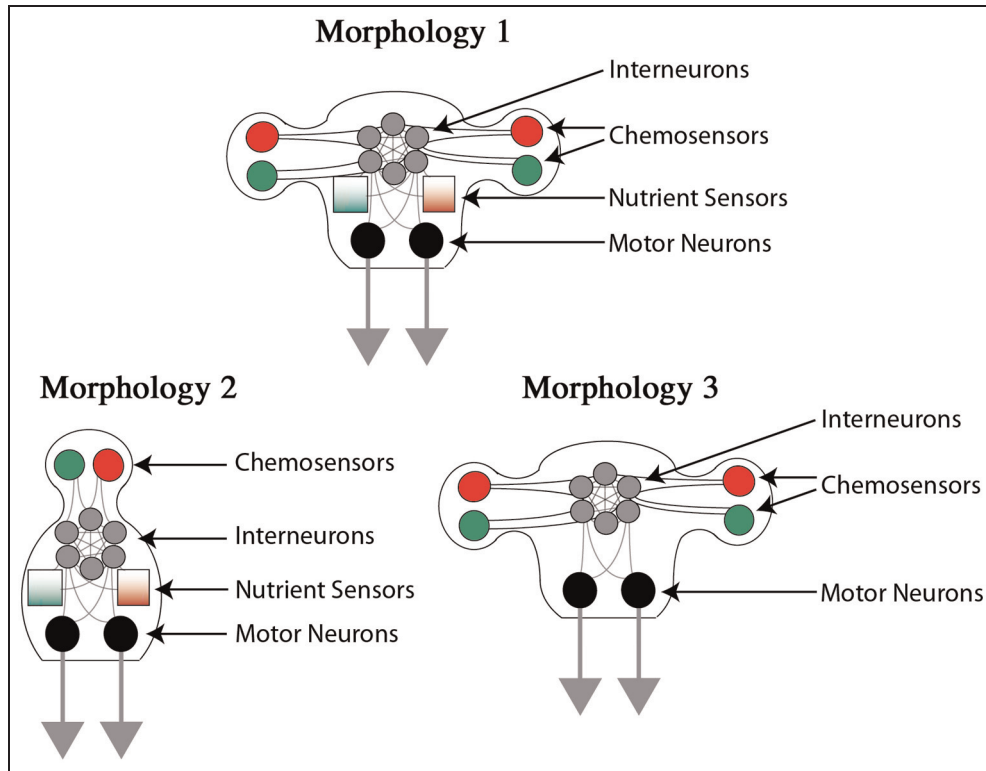


Figure 2. The three morphologies described in the experimental section. Morphology 1 has two stalks, which project outward on the two sides of its body. Morphology 2 has one chemosensory stalk, which projects outward from the front side of its body. Morphology 3 is the same as morphology 1, but has no nutrient sensors that detect the nutrient levels.

2.2 Multiple trial configurations

Each agent is tested in 11 different configurations of the environment, shown in Figure 3, and the average length of time the agent survives in these configurations defines its fitness. These particular 11 configurations were chosen to sample a variety of scenarios in order to guarantee robustness in its ability to forage successfully. Some configurations have the resources far apart, some have them close together. The agent is initiated closer to one or the other resource in different trials, and faces different directions relative to the resources. The initial nutrient levels are varied, so that the agent is more “hungry” for either nutrient in different trials.

2.3 Nervous system and genetic algorithm

The nervous system, made of the sensors, effectors, and interneurons, is modeled by a continuous-time recurrent neural network (CTRNN), and evolved with a real-valued genetic algorithm (GA). The details of the CTRNN and GA are described in the appendix.

Each individual, with a unique combination of CTRNN parameters, is put into an agent body and simulated in the 11 environments. Its parameters generate unique dynamical and behavioral tendencies. Each simulation runs to completion, either with the death of

an agent, or at the maximum time of 5000 time steps. An agent’s average longevity value becomes its fitness value, and individuals are selected from the population with rank-based selection, so that individuals with high fitness are preferentially selected. The parameters of selected individuals are mutated and are brought into the next generation. By selecting and mutating agents to maximize longevity, the GA finds increasingly successful individuals that can switch back and forth between the approaches and retrievals of the two resources.

3 Experimental results

The three morphologies, illustrated in Figure 2, are experimented upon in this section to examine the space of possible dynamical strategies. As we will see, an agent’s morphology greatly influences the resulting strategies; an agent with one chemosensor stalk must integrate the change in chemical concentrations across time as it moves, whereas an agent with two stalks can rely on the difference in chemical concentration between them. An agent with nutrient sensors can rely on this information to drive action switches, whereas an agent without nutrient sensors must somehow internalize this information if it is to succeed. In addition to the different strategies made possible by different

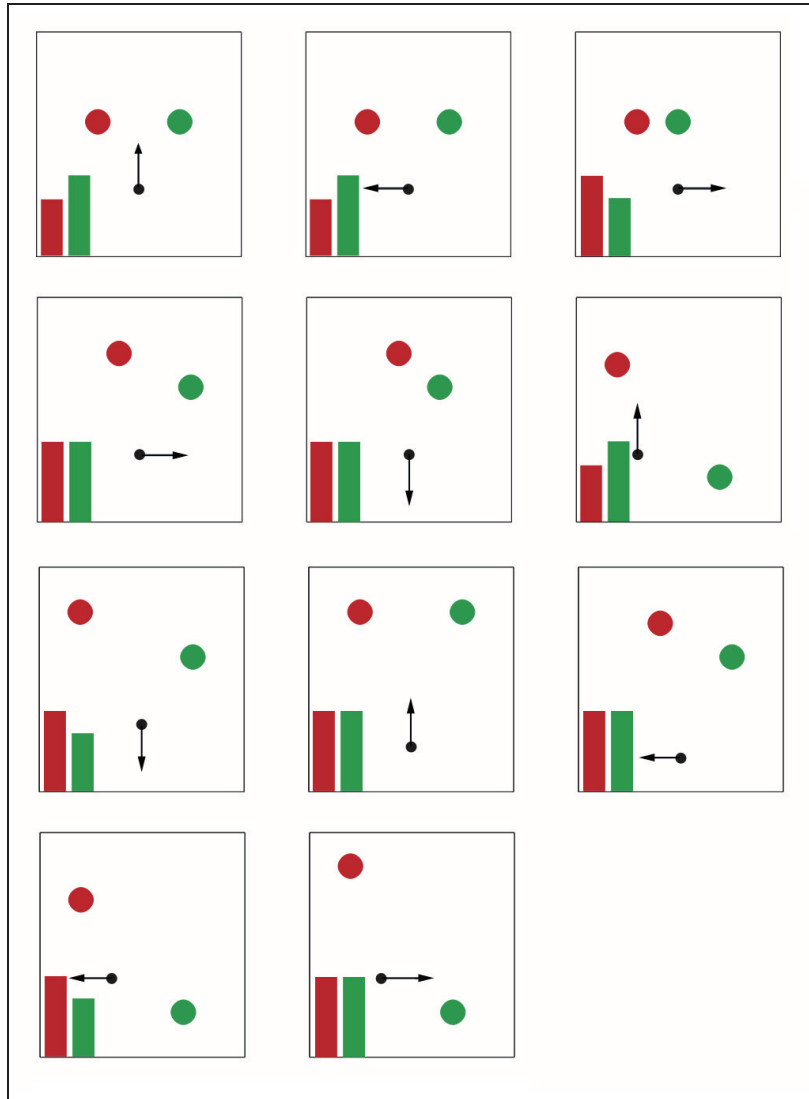


Figure 3. Eleven initial configurations. The agent is shown to be oriented in different directions, in the direction that the black line is pointing, which makes the absolute position of the resources irrelevant because the relative positions to the agent is what determines behavior. In addition to different food locations and agent orientation, the agent was initiated with different nutrient levels, which are shown as the bars on the bottom left corners.

morphologies, these experiments demonstrate how each body plan can generate multiple different strategies given its own fixed constraints.

As illustrated in Figure 2, morphology 1 has two chemosensory stalks that project outward on either side of the agent, morphology 2 has a single chemosensory stalk that projects forward, and morphology 3 is the same as morphology 1 but lacks nutrient sensors. All other morphological and experimental variables are held constant between these different morphologies. This includes the number of interneurons (six), a population size of 300, a mutation variance of 0.25, the connectivity profile between interneurons and with motor neurons, physical laws of movement, metabolism rate, eating rate, diffusion rate, and the 11 environmental conditions (with the exception of invariant

initial nutrient levels for morphology 3, this will be discussed later).

The fitness scores of individual agents are based on the amount of time survived through the 11 trials, and normalized to the maximum value of 1. A value of 1 corresponds to survival in all the trial configurations for the maximum possible time. The top-performing agents of morphologies 1, 2, and 3 achieved very similar fitness scores of 0.85, 0.83, and 0.85, respectively. This indicates that no morphology is intrinsically more limited than others, because the best agents appear to perform equally well.

However, statistical analysis suggests that the three morphologies differ in their evolvability. A one-way analysis of variance (ANOVA) was run on the distribution of final fitness values of the three morphologies.

The ANOVA rejects the hypotheses that there is no difference between the final distributions of fitness values, with $p < .05$ that they are drawn from the same population.

In order to assess the different morphologies' capacity to evolve successfully we defined a minimum fitness value and recorded the amount of evolutionary runs that led to agents above this minimum level. This number, relative to the total number of runs (which was 50 for all morphologies) determines a success ratio. The minimum fitness level was set to 0.44, which is equivalent to the normalized minimum time needed to survive through two full metabolism cycles in which the nutrient levels start full, and drop to empty. Given this definition, morphology 1 has a success ratio of 0.34, morphology 2 has a success ratio of 0.24, and morphology 3 has a success ratio of 0.24.

3.1 Morphology 1: two chemosensory stalks

This first type of body plan is provided with two chemosensory stalks. Each stalk contains both types of chemosensors, and projects outward on either side of the agent, providing the given agent with input regarding chemical concentrations at the two locations. In addition to the four chemosensory neurons, an agent with this morphology receives input from two nutrient sensors that detect how much of each nutrient the agent has at each moment.

The best performing agent has a fitness score of 0.85, which means that it survives through 85% of the time provided in the 11 trial configurations. The 15% of the time that it does not survive is spread across the four trials in which the resources are further apart. In these trials the agent still successfully engages in the behaviors, but at some point gets lost after eating one of the resources and ends up in regions of the environment with low chemical concentrations, where it cannot detect the chemical and dies before finding a resource. The best agents of the other morphologies also commonly fail these four environments. This might be because these environments are not well represented in the trails, and so they do not strongly influence the final fitness score. This failure does not matter much for the purpose of this paper, because we are interested in action switching and not in searching behavior.

Figure 4A shows a typical behavioral trajectory produced by this top-performing agent as it moves from one resource to the other and back. We see that the agent successfully chemotaxes toward both resources, and spends some time circling around and within each resource (here called "eating"). This agent has a unique strategy for eating, in which it starts with tight coils within the resource and then gradually builds up its behavior into bigger and bigger loops. The agent

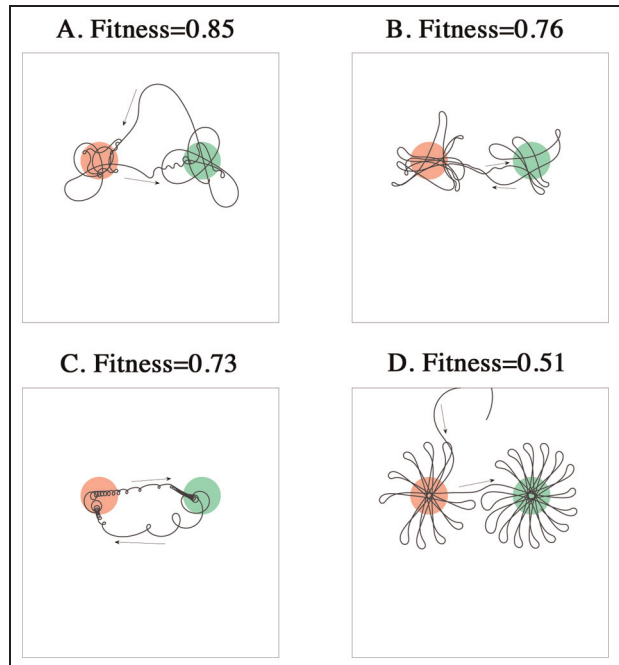


Figure 4. Four behavioral strategies evolved for morphology 1.

demonstrates successful action switching after eating, by approaching the next resource.

Unlike agent 4A, the agent shown in 4B does not change its behavioral pattern throughout the eating phase; it simply moves through the resource and after it passes the resource it turns around and comes back through. This behavior persists for a sufficient time for the nutrient to be refilled, and then it switches its action and moves towards the opposite resource. Agent 4C exhibits a variation of a common strategy, seen multiple times in other successful agents of morphology type 1, in which the agent is engaged in a constant loop-like behavior as it approaches and eats both resources. Both agents 4B and 4C exhibit robust action switching in all environments, but as with agent 4A, they occasionally get lost and die in environments with distant resources. Agent 4D, the least successful of these three, only shows robust action switching in environments with resources that are in close proximity. When the resources are far apart, the agent keeps on eating one of the resources and does not escape, thereby failing to action switch.

3.2 Morphology 2: one chemosensory stalk

The second type of morphology is provided with only one chemosensory stalk, which projects forward from its body. This requires temporal integration of the chemical signal, rather than the spatial comparison made possible by the morphology with two stalks.

The best performing agent has a fitness score of 0.83. As with the previous top-performing agent of the first morphology, this one also gets lost and dies in the same four trial environments as the agent of morphology 1,

usually after several successful action switches within those environments. The agent's behavior, shown in Figure 5A, has a simplicity that results from the fact that its left motor neuron is held fixed at its maximum level, and only the right motor neuron is modulated throughout behavior. This leads to the observed tendency to turn only towards the right. For its approach of either resource, it moves beyond the resource, where the chemical signal begins to drop. It then turns right until the detected chemical signal rises again, and then it makes a smooth, continuous turn into the resource. For its eating phases, it engages in tightening loops, until the nutrient level is resupplied. The loops loosen as the agent disengages from the eating phases, and switches to its approach of the next resource. A variation of this strategy is analyzed in depth in the next section on the dynamical analysis of action switching.

Agent 5B shows the most profound asymmetry of action observed in the agents evolved. It approaches the green resource with a persistent behavioral oscillation, which is likely used to temporally integrate the chemical signal. It then eats the green resource by engaging in a tight loop. In a very different fashion, it approaches the red resource with a straight trajectory, which is corrected after it passes the resource. It eats the red resource by first moving in straight trajectories through the resource and taking tight turns after it passes through, and then transitions to round loops. After several round loops, it switches to the approach of the green resource once again. This same agent uses a somewhat different strategy in environmental configurations where the resources are closer together (not

shown). In these environments it eats the green resource with a behavior similar to the one observed for the red resource, but the approach behaviors remain the same in these situations. Figure 5C demonstrates a strategy similar to 5A, which relies on only one motor neuron. It only turns left throughout the entire simulation, but its turns are much quicker than the strategy of 5A, and are followed by relatively straight trajectories. Figure 5D also demonstrates some asymmetry, with a straight approach towards the green resource and a winding approach towards the red resource. Additionally, the green resource is eaten by moving in tight loops, and the red resource is eaten by moving in larger loops. This agent gets lost very easily in most of the environments, but usually after several successful rounds of eating.

3.3 Morphology 3: two chemosensory stalks and no nutrient sensors

Morphology 3 has the same body plan as morphology 1, but without nutrient sensors to provide input regarding nutrient levels. Additionally, the 11 trials are slightly modified. For the previous two morphologies, there were varying amounts of nutrient levels at the onset of the trials. This would be a limiting constraint in morphology 3's evolution, because it does not have the means with which to tell the difference between different nutrient levels, and would therefore be penalized for something it could not prevent. To correct for this, all nutrient levels are initiated at the full level across the 11 trials.

This morphology is of considerable interest because successful agents must make use of a strategy that determines how long to eat, and the appropriate resource to approach, without using nutrient level as an informational measure for influencing behavior. The previous two morphologies could rely on nutrient levels to determine when they had sufficiently filled their nutrient supply, and they could determine which resource to approach based on what nutrient supply was lower. But successful agents of this third type have to use some form of memory to determine how long they eat, and to guide them from one resource to the other without accidentally coming back to the already-gathered resource.

The top-performing agent has a fitness score of 0.85, which means that it is very robust, though it eventually dies in the same four trial configurations as the other top agents. The behavior shown in Figure 6A illustrates its strategy: it approaches a resource, slows down as it moves across it, then it turns around, comes back over the resource, and leaves for the next resource. This behavior is considerably more ordered than the behaviors of previous morphologies. The fixed behavioral pattern likely serves as the form of memory

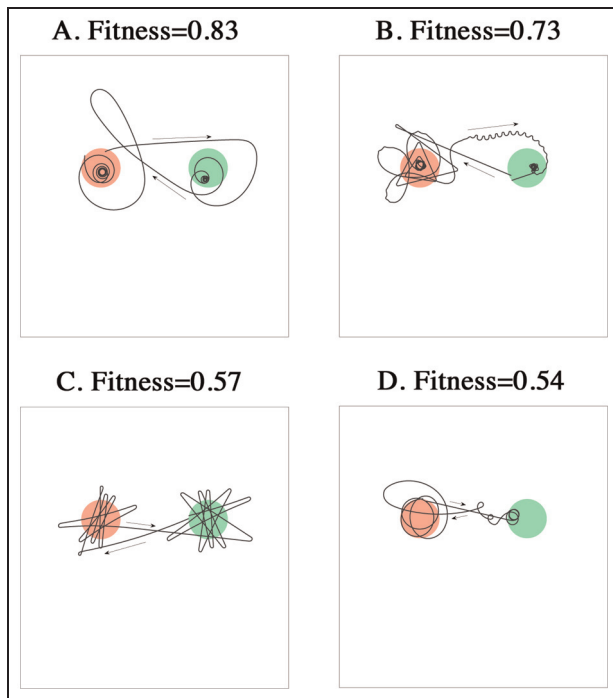


Figure 5. Four behavioral strategies evolved for morphology 2.

hypothesized in the previous paragraph. Rather than relying on nutrient information, it relies on highly ordered motor output to determine how long it should eat. It never comes back to the same resource that was just eaten, possibly because it engages in a relatively straight trajectory away from the resource after the action switch. In some of the environmental configurations not shown here, the straight trajectory is not oriented towards the opposite resource, and it is able to successfully correct its direction.

Agent 6B has a fixed behavioral pattern during the eating phase, but with an additional asymmetry between the eating of the two resources. In the illustrated behavior, it eats the green resource with three elongated loops, and eats the red resource with four tighter, rounder loops. This strategy is fixed for the entire duration of a simulation given this particular environment, but it varies between environments. When the resources are closer together, the agent engages in fewer loops on top of the red resource, and in more loops on top of the red resource when resources are further apart. Its eating behavior of the green resource always looks the same. Agent 6C has a fixed looping behavior as it crosses either resource, but this also changes between environmental configurations; when the resources are closer together it engages in fewer loops before switching to the next approach. Agent 6D engages in four broad loops at every eating stage across all environments, which also provides a fixed time interval that sufficiently fills the resource supply. This agent sometimes moves back to the same resource that was just eaten, leading to its death.

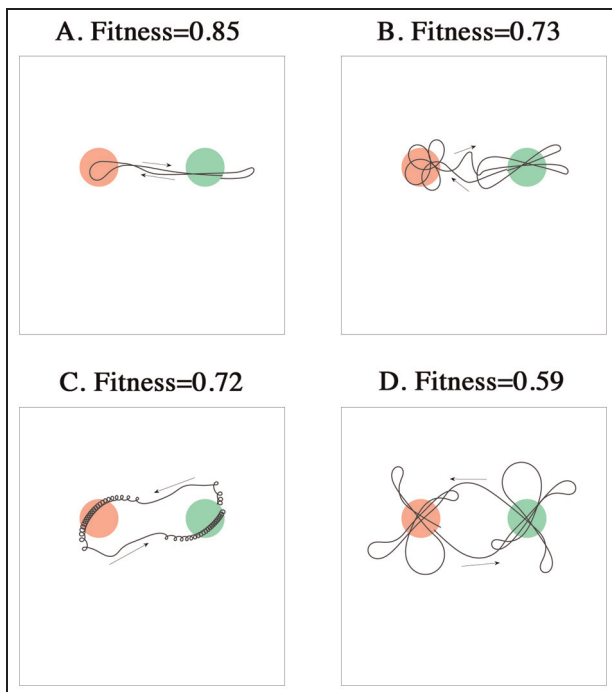


Figure 6. Four behavioral strategies evolved for morphology 3.

4 Dynamical analysis of an action switcher

In this section, we extend the previous results by selecting a single agent and fully analyzing the dynamical properties that underlie its behavior. For the purpose of producing a simplified dynamical analysis, we used the body plan of morphology 2 and reduced its interneuron count to four total interneurons, bringing its total number of neurons to 10. Four interneurons is the minimum number of interneurons that were able to generate successful behavior when optimized with GAs. Multiple agents evolved successfully, but at a success ratio of 0.20, which is lower than the evolutionary runs for the 12-neuron morphology 2. The top-performing agent of this reduced type, whose behavior is shown in Figure 7, had a fitness score of 0.84, which is slightly higher than the previous agent shown in Figure 5A. This indicates high robustness in the 11 environmental configurations.

The agent arrived at a behavioral strategy similar to the top-performing agent of morphology type 2, in which only the right motor neuron is actively modulated and the left motor neuron is held constant at its maximum level, leading to a rightward-turning inclination. This fact is taken advantage of for our analysis because we can reduce the behavior's dimensionality by using the right motor neuron's activity alone and ignoring the left motor neuron.

There is a straightforward way of breaking up the behavior qualitatively in a way that later will become quantitative; we assume that the agent's behavior is divided into two approach actions towards either resource, and two eating actions. These are shown by the different coloration in the behavioral trajectory of Figure 7, with the two approach phases shown by the solid red and green lines and the eating phases shown by the dashed red and green lines. In the approach phases, the agent moves relatively straight towards a resource, and then overshoots the resource. After the overshoot, it begins to turn right, and slowly turns into a tighter curve as it approaches the resource. When the agent crosses the resource boundary, the corresponding nutrient level begins to increase. This initiates an eating phase. During the eating phases, it engages in ever-tightening loops within the resource. This behavior abruptly ends, and after a short transient during which the agent reorients itself, the agent proceeds to move towards the next resource.

It should be noted that an intuitive division of the behavior might not lead to this exact segmentation, but with a dynamical analysis we found that there is a clear, quantitative method to justify these divisions. In the following analysis, we will see that these four phases of the behavior are neatly divided into distinct dynamical pictures, and appear to follow specific rules that can be derived from the dynamics. These individual segments will interchangeably be called sensorimotor coordination modes or behavioral modes, and the rules that

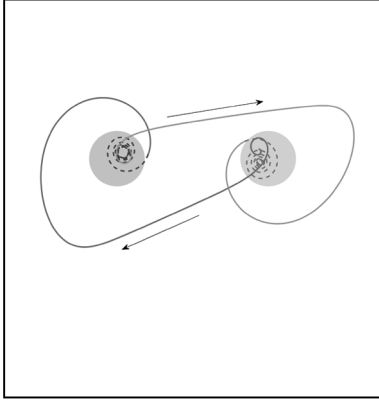


Figure 7. Behavioral trajectory of simple action switcher.

guide them will be called sensorimotor coordination surfaces. With the introduction of the sensorimotor surfaces in Section 4.3, the significance of this particular division will become more obvious.

4.1 Global analysis of agent's neuronal behavior

We begin our analysis with a reduced phase portrait of the agent's nervous system through its entire global behavior, shown in Figure 8. This takes the internal perspective of the nervous system, looking only at the CTRNN's states as the agent moves through all four of the local behaviors, without accounting for the environment. This will be followed in the next two sections with a more local examination of the individual behaviors as generated by the CTRNN coupled to a body and environment.

The agent's neuronal dynamics exists in a 10 dimensional state space made up of the states of the two chemosensors, two nutrient sensors, two effectors, and four interneurons. To produce a visualization of the neurons' trajectory through this space, the dimensionality has to be reduced. This is accomplished by performing a principal component analysis (PCA) of the observed neuronal trajectory, and converting the original trajectory into a coordinate space defined by the top two principal components. We simulated the agent as it moved from one resource to another and back (in the same trajectory shown in Figure 7), and recorded all of the neurons' states throughout this behavior. PCA found that the top two principle components account for approximately 91% of the variance in this typical neuronal behavior. (When a longer trajectory, composed of many full rounds of action switching between the resources is analyzed with PCA, the top two principal components account for 84% of the variance. This reduction is due to the fact that the CTRNN does not move through the same exact trajectory with every action.)

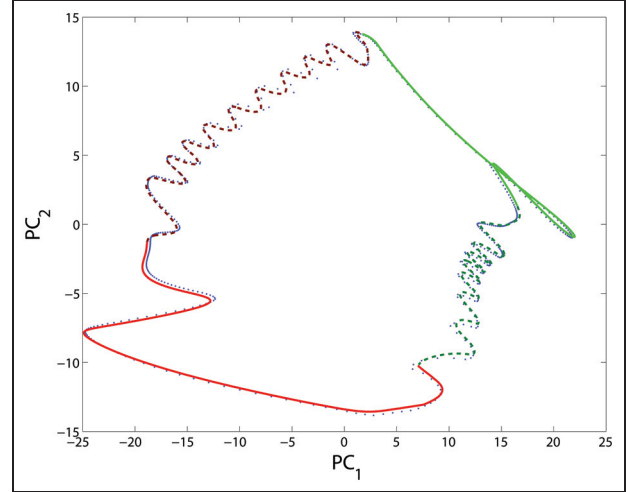


Figure 8. Neuronal dynamics for the behavior shown in Figure 7. The neurons' trajectory through their state space is shown in the four different colors that designate different behavioral modes. This trajectory is projected onto the top two principal components. The blue dots are the single equilibrium point shown in different locations as it moves throughout a behavior.

We examined the amount that each state contributes to the first principal component, which accounts for approximately 78% of the variance (59% when the extended trajectory is analyzed). Much of this variance comes from two of the interneurons, interneurons 1 and 4, indicating that their activity contributes significantly to the neuronal dynamics. Most of the remaining variance comes from the sensor states, which are nicely balanced in their contribution to the first principal component; where one chemosensor is pointing in a positive magnitude of 0.18, the opposite chemosensor is pointing in the negative magnitude of -0.16 . The same pattern holds for the nutrient sensors; one is -0.13 , the other is 0.14. This demonstrates that the relationship between the sensors is negative: when one is high the other is low.

The trajectory of Figure 8 shows the neurons' states as they unfold through the coordinate space defined by the top two principal components. The coloring scheme corresponds to the segmentation of Figure 7, but rather than looking at the agent's movement within the environment, this trajectory illustrates the neuronal states throughout each of the four modes. We see some interesting parallels between the neuronal states and the observed behavior. During the two eating phases, shown as dashed red and green lines, there is an oscillating trajectory, which corresponds to the fluctuating chemosensory input that results from the agent moving in loops through the chemical gradient. The approach phases, shown by the solid red and green lines, have sections that line up with the behavioral overshoot of

the resource, and then the gradual rightward turn into the resource.

To expose the dynamical properties that guide this trajectory, we recorded the agent's sensory inputs through this entire simulation, and calculated the system's equilibrium points (EPs) given those sensory values. The EPs are projected onto the transformed PCA coordinate space, and are shown as the blue dots in Figure 8. It was discovered that the system only contains a single stable EP for the entire duration of the simulation. We were surprised to find this rich behavior arising from a single EP that moves through the agent's internal state space as sensory states are changed; originally, we expected a bifurcation to produce the sudden transitions between actions.

It is apparent in this figure that the neuronal trajectory follows the EP's trajectory very closely. It is never able to reach the EP, because as it approaches the EP, the motor neuron is active and produces output. When this output is translated into behavior within an environment, the sensors' values change and the EP moves, leaving the EP forever just out of the present state's reach.

4.2 Experiments on local sensorimotor behavior

The previous section provides us with an understanding of how the system unfolds through its state space by chasing an EP. But due to the transformation of the axes with PCA and the reduction of dimensionality, we lose sight of how particular actions unfold in a situated manner.

In this section and the next, we aim to bring the relationship between sensors and effectors back into focus, and to uncover how the agent produces successful behavior in the environment. The simple action switcher analyzed here was chosen in part because of its strategy that relies on only one motor neuron's active modulation. This fact allows us to study the sensorimotor system by analyzing how different sensory inputs affect the single motor output, and through their relationships determine the various actions. There are four sensory inputs: the two chemosensory inputs and the two nutrient sensory inputs. We study how these inputs affect the motor output by investigating the state space that includes the agent's set of possible sensory states, and how these states influence the single EP as projected onto the motor output. But rather than create a five-dimensional phase portrait that captures all the different actions, we aim to study each behavior separately and identify a reduced number of variables that are relevant for each action.

Several experiments were first run to uncover the relevant variables for each action. Figure 9 displays a baseline condition and five experimental conditions. Figure 9A shows the baseline condition, in which the agent is initiated in the environments with no

constraints on the sensory inputs. It moves through two complete approach and eating cycles, which takes approximately 2300 time steps. To keep the conditions constant, the same amount of time steps are used in the rest of the illustrated experimental conditions. In each experiment, we target individual sensors and hold them fixed throughout a simulation in order to identify what sensors matter for each action. Figure 9B holds the green nutrient sensor, N_{green} , at a value of 0. Figure 9C has the opposite condition, with $N_{\text{red}} = 0$. These demonstrate that when a single nutrient sensor is held fixed at a low level, the agent moves towards the corresponding resource and remains on that resource for the remainder of the simulation. It loses its ability to switch to the other resource, as if it remains forever hungry for the nutrient that is held low.

The next three experiments, shown in Figures 9D through 9F, demonstrate that when the agent is approaching a resource, the opposite nutrient sensor needs to be modulated. Figure 9D shows that if both nutrient sensors are held at 0, $N_{\text{green}} = N_{\text{red}} = 0$, then the agent does not approach any of the resources, but instead circles around in a tight loop at its initial position. The same behavior is seen when N_{red} is held at 0, and N_{green} is held at any value between 0 and 6.9. When N_{green} is held approximately between the values of 6.9 and 8.1 there is a destabilization of the behavior, shown in Figure 9E in which $N_{\text{green}} = 8$ and $N_{\text{red}} = 0$. When N_{green} is held at values above 8.1, the agent moves towards the red resources, but circles it from a distance. This is shown in Figure 9F, in which $N_{\text{green}} = 8.4$ and $N_{\text{red}} = 0$.

These experiments indicate that when the opposite nutrient sensor is held fixed, the agent loses its ability to approach a resource. The opposite nutrient sensor's modulation guides the approach towards a resource; in order to approach the red resource, the red nutrient sensor has to be low, and the green nutrient sensor has to transition from a high value to a low value. This guides the agent from moving towards the resource at first to a tight loop behavior into the resource. The same pattern has been confirmed for the approach of the green resource.

Other experiments, not illustrated here, demonstrated that the opposite chemosensor also does not matter for the approach behaviors and that neither of the chemosensors matters for eating behaviors. In the first set of these experiments, we held both the nutrient sensor and the opposite chemosensor fixed at low values, and the agent still approached each of the resources in approximately the same trajectories as those shown in Figures 9B and 9C. This means that a reduced subset of sensory states is relevant for the different approach behaviors; N_{green} and CS_{red} are the sensors that matter for the approach of the red resource, and N_{red} and CS_{green} are important for the approach of the green resource.

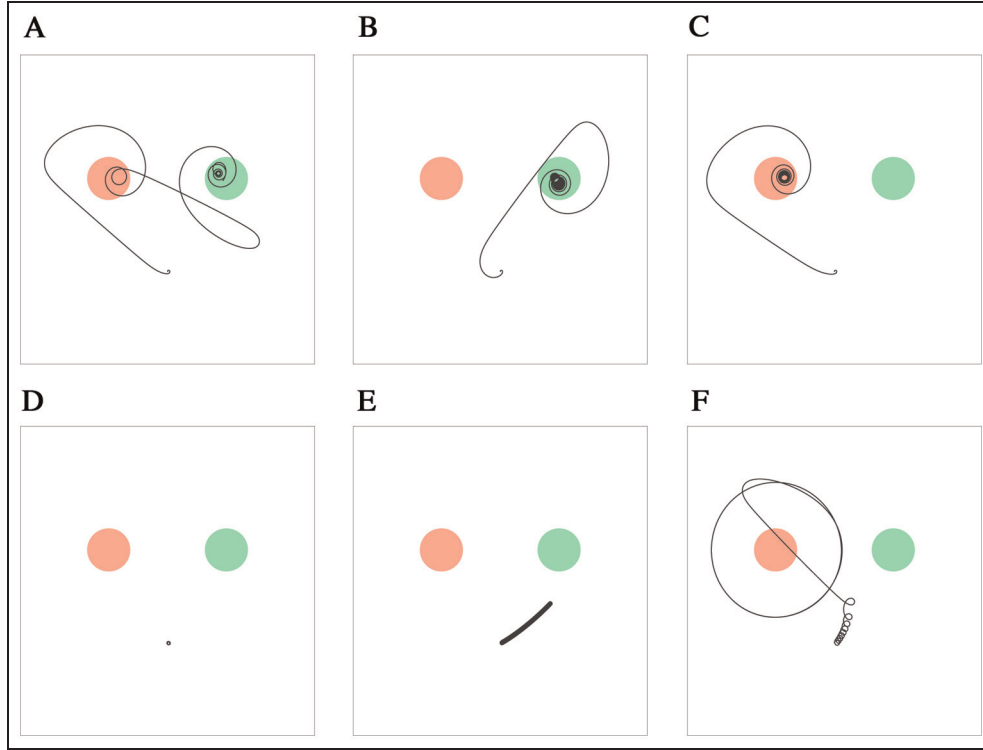


Figure 9. (A) Baseline condition with all sensors allowed to vary. (B) Experimental condition with one fixed sensor: $N_{\text{green}}=0$. (C) One fixed sensor: $N_{\text{red}}=0$. (D) Two fixed sensors: $N_{\text{red}}=0$, $N_{\text{green}}=0$. (E) Two fixed sensors: $N_{\text{red}}=0$, $N_{\text{green}}=8$. (F) Two fixed sensors: $N_{\text{red}}=0$, $N_{\text{green}}=8.4$.

In a second set of experiments not illustrated here, we began the simulation with baseline conditions and fixed both chemosensors only during the eating behaviors; the experiment is initiated as soon as the agent moves within the boundaries of a resource. In these conditions, the agent continues eating with no major effect from the experimental manipulation. This demonstrates that the modulation of chemosensors, CS_{green} and CS_{red} , is not required for either of the eating behaviors. This is likely due to the fact that during the baseline condition's eating phases the agent remains in relatively fixed positions relative to the resources.

Figure 10 illustrates the interneuron gating mechanism that allows certain sensors to drop out (or be ignored) during the approach behaviors. Interneurons 1 and 4, which are the same two interneurons that contribute a significant amount of variability to the first principal component, have evolved to act as gates of information between the sensors to the right motor neuron. By acting as a gate, we mean that some information is allowed to pass through these interneurons and influence behavior while other information is blocked or ignored. In the illustrated example, we show how CS_{green} and N_{red} drop out during the approach of the red resource and have no effect on the interneurons' activity, while the modulation of CS_{red} and N_{green} has a strong effect on the interneuron states.

The black lines in the plots are the interneurons' stable state output given their full range of total input. The markers imposed on top of this stable output are the activations of the interneurons with N_{red} held fixed at 0. The blue asterisks show the effect on interneuron activity due to full chemosensory modulation (chemosensory modulation means that both chemosensors are tested at values between 0 and 10) when $N_{\text{green}}=10$, and the red circles show the effect of full chemosensory modulation when $N_{\text{green}}=0$. CS_{red} contributes mostly to the input of interneuron 1 and CS_{green} contributes mostly to the input of interneuron 4. This is due to CS_{red} having a high connection weight of 9.74 to interneuron 1 and a low weight of -0.42 to interneuron 4, and the opposite for CS_{green} , with a low weight of -0.29 to interneuron 1 and a high weight of -12.86 to interneuron 4.

To summarize the findings of this experiment; the sigmoidal nonlinearity of the interneurons is used to remove the effects of some sensors, and to enhance the effects of others. The graphs of Figure 10 show that when the red nutrient level is held at 0, and the green nutrient level begins at level 10 and drops to level 0 (which is approximately the case during the approach of the red resource), the red chemosensor greatly modulates the output of interneuron 1 and the green chemoreceptor has no effect on the output of interneuron 4 due to the interneuron's saturation at a low output value.

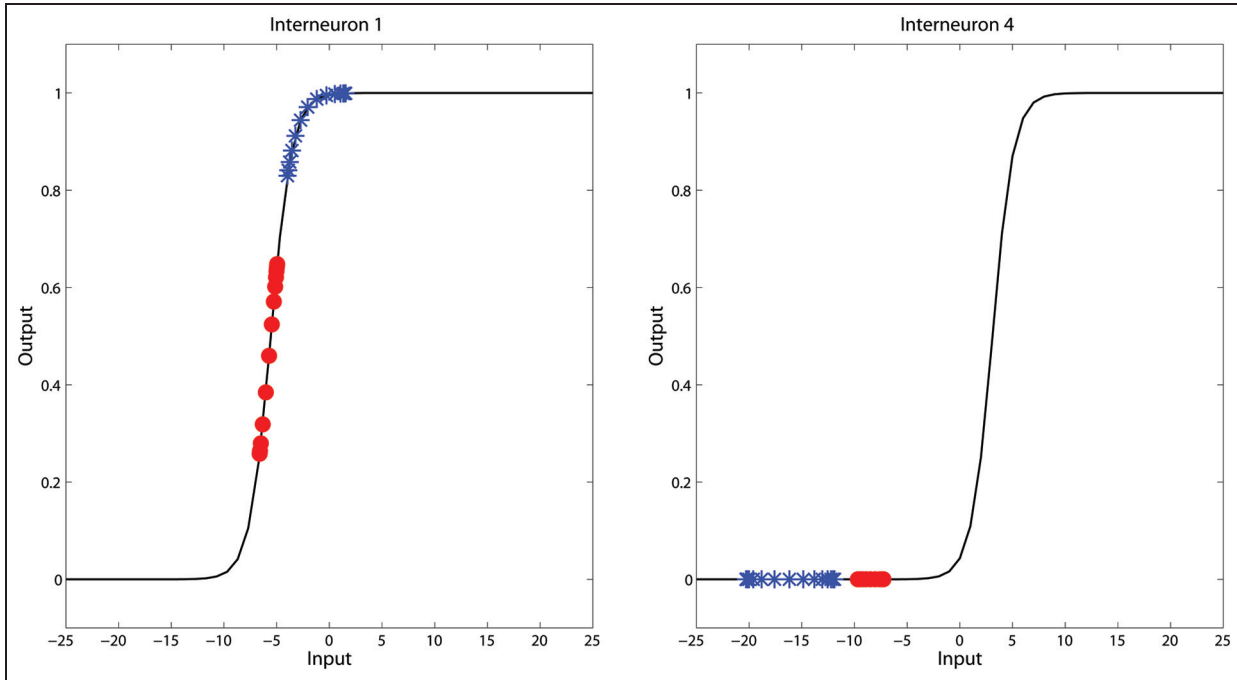


Figure 10. The black lines show the two interneurons' stable state output given the range of input. The red circles and blue asterisks are the activations of the interneurons when $N_{\text{red}}=0$. The blue asterisks show the effect of full chemosensor modulation when $N_{\text{green}}=10$, and the red circles show the effect of full chemosensor modulation when $N_{\text{green}}=0$.

An analogous plot not included here would show that the exact opposite is true when the green nutrient level is held at 0 and the red nutrient level moves from 10 to 0, with the green chemosensor's modulation strongly influencing the output of interneuron 4.

In summary, there are two types of mechanisms that allow certain sensors to drop out in different behavioral modes, and give control to the relevant sensors. During the eating modes, in which the agent is engaged in tight loops on top of a resource, the chemosensor states do not vary much because the agent is not moving very much spatially. This gives control to the nutrient sensors. During the approach modes, the interneuron gating mechanism blocks a pair of nutrient sensor and chemosensor such that their modulation has no effect on the motor neurons. So the distinction here is between the cases in which sensors are dropped out due to a situated behavioral condition versus the cases in which sensors are dropped out due to internal gating-like mechanisms.

4.3 Sensorimotor coordination surfaces

These experimental findings suggest that there are local rules of coordinated behavior that guide individual actions. This prompted the search for, and discovery, of what we call *sensorimotor coordination surfaces*, which are shown in Figure 11. These are approximate local control surfaces that define individual actions, and greatly reduce the dimensionality of the overall

state space. Each behavioral mode has its own surface, which is made of the system's EPs given the restricted set of sensory values shown in the x - and z -axes, and projected onto the motor output value of this EP, shown on the y -axis. In these graphs, nutrient sensors are labeled N , chemosensors are labeled CS , and the motor neuron is labeled M . For each surface, the two sensors that are not included are held at fixed values. For surface A: $CS_{\text{green}}=0$, $N_{\text{red}}=0$, for B: $CS_{\text{red}}=2$, $CS_{\text{green}}=9.5$, for C: $CS_{\text{red}}=0$, $N_{\text{green}}=0$, and finally, for D: $CS_{\text{red}}=9.5$, $CS_{\text{green}}=2$. By holding these values fixed, only changes in the relevant sensors can affect the surface's landscape.

The trajectories of the individual coordination modes are projected onto this graph, with each mode's trajectory matched with a single unique sensorimotor surface. Visually, they appear to follow the topography defined by the surfaces. These trajectories come from the same simulation as Figures 7 and 8, and correspond to the same color code. The close contact between the trajectories to the surfaces further indicates that for each of these coordination modes, the given subset of sensors shown in the x - and z -axes dominate in control over the motor output. This close contact is further demonstrated in Figure 12, which shows vertical distance from the four surfaces across time. During each behavior, the trajectory approaches one of the surfaces and when the behavior ends, the trajectory departs from the given surface and approaches a different one.

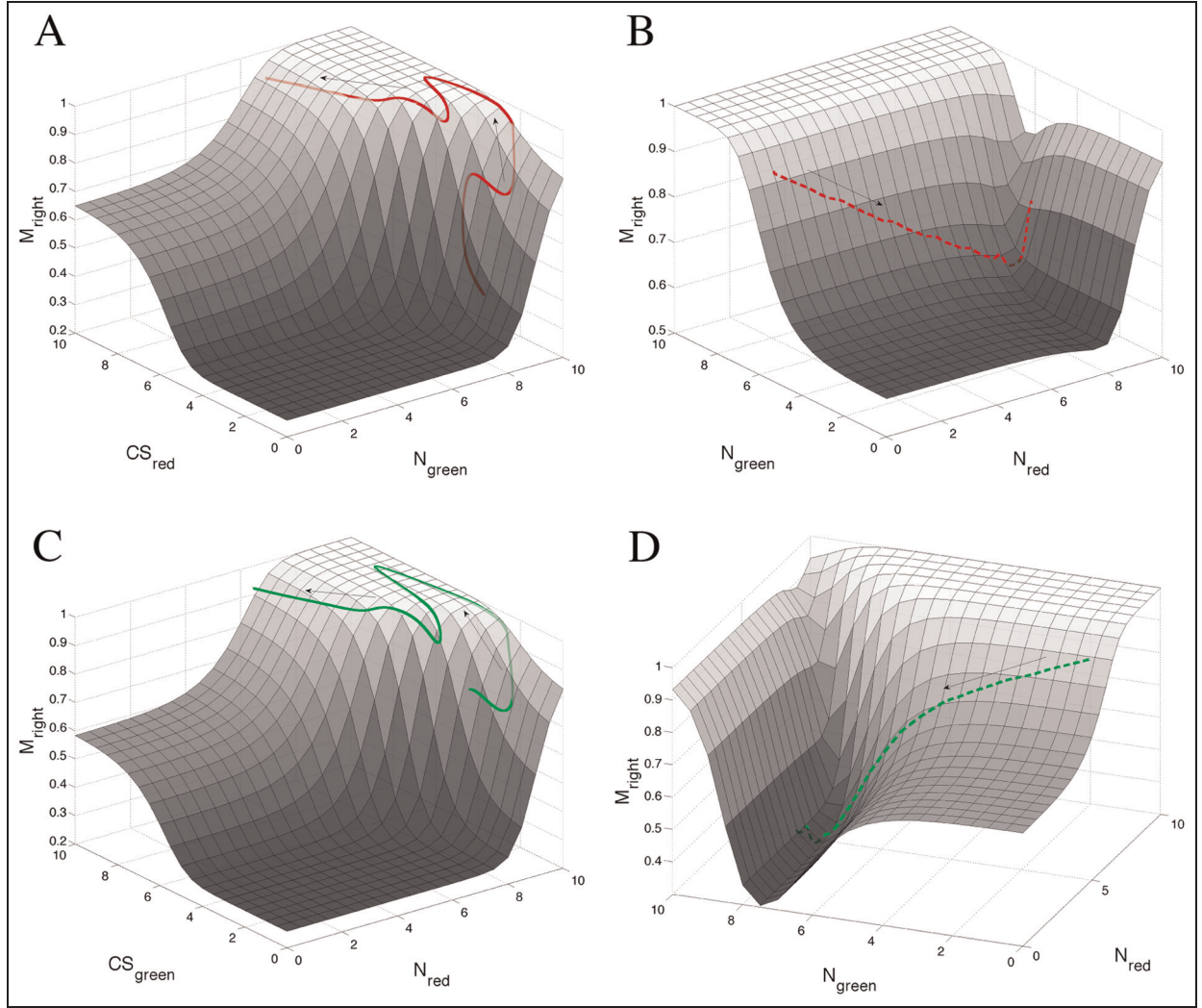


Figure 11. Sensorimotor coordination surfaces. These surfaces show the stable equilibrium value of the motor output given that the two sensors on the x- and z-axes are held at their given values, while the other two sensors are held fixed. Simulated trajectories of four different behavioral modes, shown with the same coloration in Figures 7 and 8, are brought into the same space to demonstrate their close fit with the surfaces.

The surfaces in Figures 11B and 11D define the red and green eating behaviors, respectively. Only the two nutrient sensor states are relevant for the observed outputs, which, as mentioned in Section 4.2, is because the chemosensor states are relatively fixed during the eating actions. For these eating behaviors, as one of the nutrient levels rise due to eating, this pulls the system's state across the surface and towards an EP with much lower motor output. This reduced motor output brings the agent into the ever-tightening loop shown in Figure 7. At the end of the eating actions the unfolding trajectory comes upon a sharp rise in the surface, leading to a straighter behavioral direction within the environment; this is what drives the agent out of the looping behavior and into the next action.

The surfaces in Figures 11A and 11C define the two approach actions. The relevant sensors, as demonstrated in Section 4.2, are the chemosensor that detects

the resource being pursued and the opposite nutrient sensor. As the agent approaches a resource, the corresponding chemosensor state increases. The agent overshoots the resource in both of these observed actions because it keeps on moving past the resource due to its straight behavioral trajectory. Because of these overshoots, the chemosensor state drops, which moves the trajectory across the sensorimotor surface towards a region that has a lower motor output value. This makes the agent turn towards the right. After turning rightwards, the chemosensor level increases due to the agent having turned sufficiently towards the resource. This increase in the chemosensory level moves the trajectory through the surface towards a region in which the motor output is higher, leading to forward movement towards the resource. The agent continues to approach the resource with a slight curve determined by the sensorimotor surface's shape.

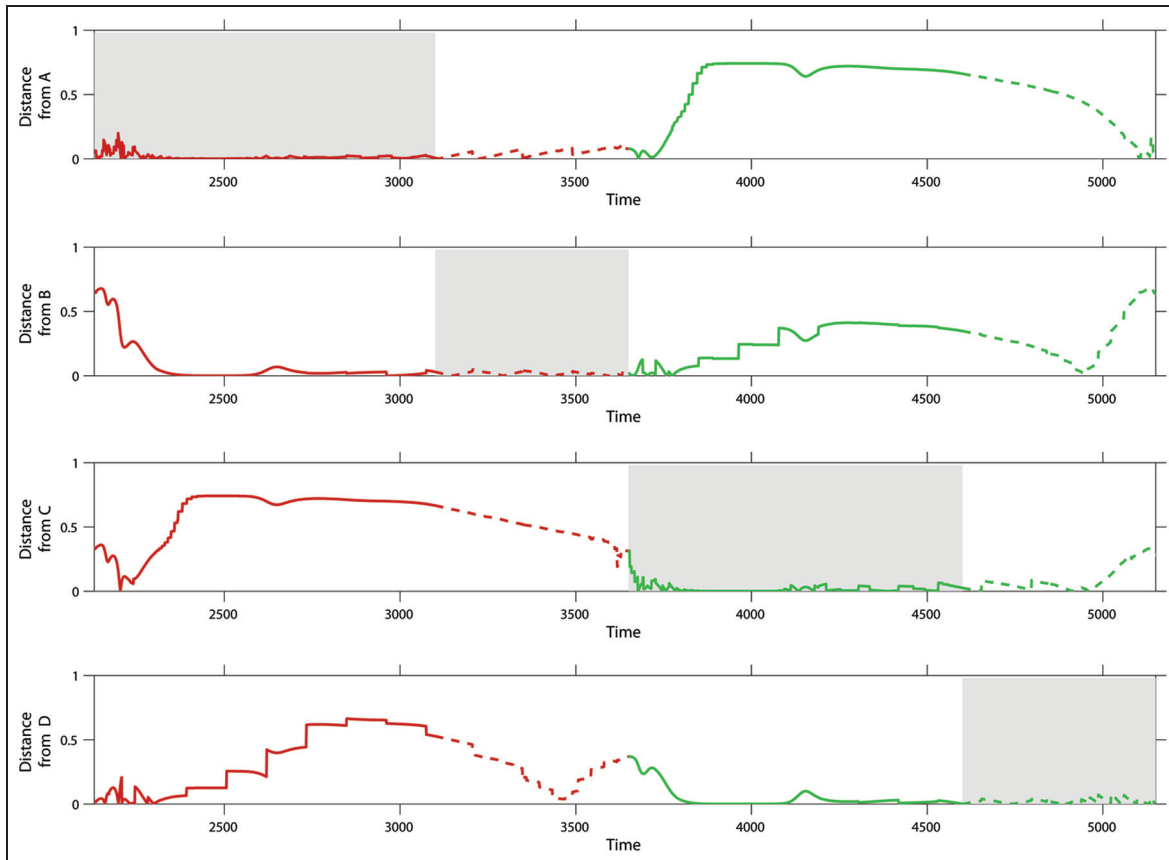


Figure 12. Distance from the four sensorimotor coordination surfaces across time. These plots begin with the approach of the red resource, and then show eating the red resource, approaching the green resource, and ends with eating the green resource. The time in which the agent engages in each surface's given behavior is highlighted in grey. The top graph shows distance from surface A, and it illustrates that the trajectory in state space comes close to surface A during the approach of the red resource, which is the behavior that surface A has been said to control. The eating of the red resource also remains close to surface A, but it is relatively closer to surface B.

The graphs in Figure 12 show the distance of the state trajectory from each of the four sensorimotor surfaces throughout the entire simulation. These demonstrate that each behavior approaches a particular sensorimotor surface for the duration of the behavior. It is also clear from these graphs that the trajectory during the approach of a resource remains relatively close to the surface defined for eating, and vice versa. This might suggest that there is no clear qualitative difference between approach and eating behaviors, but if one looks more closely at the trajectories there do appear to be some deviations from the opposite surface that are better captured by the behavior's defined surface.

These four behavioral modes appear to unfold within subspaces of reduced dimensionality as defined by the four sensorimotor surfaces, but it is important to remember that they still exist within the system's higher-dimensional state space. The sensorimotor surfaces are simplifications that provide an approximation of individual actions' sensorimotor strategies. While

they nicely divide the actions, there are some subtleties of the dynamical landscape that are missed as a result of the reduction. Each sensorimotor surface's fixed sensory variables do in fact vary throughout a given action's behavior. This variation likely does have a slight effect on the EP and its corresponding motor output, though this effect is greatly overpowered by the dominant sensors.

As for switches between actions, they seem to disappear from our explanation. Instead of discrete actions that are selected by a higher-level mechanism, we find continuous dynamics that can be broken up into dynamical regions of temporarily reduced dimensionality, with fast transitions in which the agent's dynamics move between these regions. This gives the illusion of action switching, in which different sensors appear to coordinate directed actions and when a certain goal state is achieved, the previously relevant sensorimotor coordination surface is no longer a good approximation and gives way to a new relevant sensorimotor coordination surface.

5 Discussion

We have studied the behavioral dynamics of action switching both with a coarse overview, in the many behaviors produced by different morphologies, and with a detailed dynamical analysis of a particular agent of morphology type 2. With the detailed analysis of the action-switching agent, we took a broad look at the dynamics of its entire nervous system's state space throughout the multiple actions, and at the local dynamics of specific actions and how they generate the different actions. We then described how these pieces fit together through the different switching mechanisms. This provides a clear, quantitative illustration of how we can think of multiple actions and the transitions between them self-organizing in embodied dynamical agents.

Here, actions are shown to be local dynamical properties of a brain, body, and environment, in which a subset of sensors and effectors are temporarily engaged in unique modes of coordination, defined by unique sub-surfaces of the overall dynamical landscape. These sensorimotor surfaces are a dimensionally reduced approximation of the total system's dynamics, but they are good approximations because they describe the constrained dynamics that allow the system to coordinate movement within the environment towards what appear as goals (such as reaching a resource, or filling up a nutrient supply). When an action is completed and the goal state is reached, the same dynamical processes that brought the action into being transition to a different action by engaging a different subset of sensors and effectors in a different mode of coordination.

This is a very different picture from the one provided by traditional approaches to action selection, in which an agent retrieves information from all of the sensors, passes this information up a hierarchy, compares the available information with internal information on the agent's values and needs, computes the most profitable action through a process of arbitration, and then initiates the action mode by setting the appropriate initial conditions. What the agent does in the illustrated dynamical example is more integrated and arguably simpler. Instead of having to compute a solution and evaluate its possible options, it just engages the relevant sensors and ignores the irrelevant sensors, passing the relevant information through the interneurons to the effectors. In this transfer between sensors and effectors, the dynamical information is constrained in a law-like way that is defined by the local sensorimotor surfaces.

This dynamical description shares many of the same properties put forward by the sensorimotor contingency (SMC) approach, which argues that perception and action are constituted by lawful regularities of sensorimotor flow. There is a clear connection to recently published work that uses dynamical systems to ground the theory of sensorimotor contingency in formal models (Buhrman, Di Paolo, & Barandiaran, 2013). In this

recent paper, four distinct types of SMCs are described, of which "sensorimotor coordination" is the best fit to the dynamical strategy analyzed in the present paper.

Rather than using a higher-level mechanism to select between possible actions, these agents use nonlinear dynamical strategies to block the irrelevant sensory information, accentuate the relevant information, and direct the sensors' dynamical influence on the appropriate effectors. We demonstrated two different dynamical mechanisms that allow some sensors to be engaged, and others ignored. The eating actions make use of situated invariants, in which the chemosensors are held relatively fixed throughout the behavior because of the particular situated context, and so they can be disregarded when analyzing the particular mode of coordination. The approach behaviors arise through interneuron gating, which uses the nonlinear properties of neuronal saturation to negate the effects of some sensors' activity, and engage the relevant sensors. These two mechanisms likely generalize to other architectures of Section 3, because some sensory information always needs to be ignored and other information enhanced in order to accomplish the action-switching task. The demonstrated mechanisms seem like straightforward and generalizable dynamical strategies that can accomplish this. There are perhaps other strategies not yet discovered.

We gain an understanding of the nervous system as existing in a high-dimensional state space made up of all of its neurons' states. All of the system's behaviors result from the internal state coupled to the environment, which drives the system through its state space in a coordinated manner. Specific actions, which result from the agent's neuronal state being coupled to an environment through its body, generate a transient feedback process that defines dimensionally reduced coordination surfaces. Again, it should be stressed that these surfaces are not real components of the system, but are practical modeling methods that can be used to approximate the agent's local behavior towards a goal state. With this perspective, action switching becomes rapid transitions between different subspaces of the total state space, in which some of the relevant sensors and effectors disengage and then others engage to define a new sensorimotor coordination surface.

The analyzed agent demonstrated just one strategy out of many possible strategies, some of which were shown in Section 3. Each of these might have found a different approach to solving the provided task, and so the details of this dynamical analysis are not expected to generalize to all of these different agents. But we do expect the ideas of local sensorimotor surfaces for defining particular actions to extend, along with similar switching mechanisms that define how information passes through the system. The point of the analysis was to demonstrate possible nonlinear dynamics underlying action switching in interactions between brain, body, and environment.

It is our hope that this exploration of basic dynamical properties will lend itself to the development of a more elaborate framework for the dynamical approach. The issues described here can be greatly extended and tested by scaling up the simulations to more complex agents that engage in an increased number of actions. Real living organisms have many more degrees of freedom that need to be coordinated, and more degrees of freedom in the environment that can be observed to determine context. With this scaling in complexity, dynamical agents need to develop different techniques for identifying relevant sensory information and freezing the appropriate degrees of freedom. The current findings provide preliminary methods for defining particular actions and identifying their dynamical properties, which might extend to descriptions of more complex actions.

In the future, this discussion should also be extended to different types of switching tasks. In this paper we showed a case of having different sensors with the same effectors, but we can also investigate tasks that require the same sensors and different effectors (as in Williams & Beer, 2013), or different sensors and different effectors. There can be overlapping subsets of either sensors or effectors, such that the different actions require some of the same sensors and effectors, but not all. There can also be cases in which the same sets of sensors and effectors are engaged, but in completely different patterns (as is done in Izquierdo & Buhrmann, 2008, but there the switch is induced when the modeler explicitly changes the agent's morphology). Another point that needs to be addressed in future research is that living systems can be said to engage in multiple different actions at once, such that the nervous system has to generate different coordination modes at the same time and minimize interference between them. It would be interesting to fully address all of these possibilities, and to describe the different dynamical processes that might underlie them.

An additional point of interest that should be addressed in future research is the development of skillful actions through experience. Real living systems are not born with their full behavioral repertoire, as are the agents shown in this paper. Instead, they develop many of their actions through processes that refine their behaviors to fit an environment. This adaptive capacity requires that as they dynamically unfold both in their internal states and through their coupling with an environment, they enter previously unexplored regions of their state space that engage the novel behaviors. We would expect law-like coordinated behaviors to gradually arise where earlier there was no coordination, and perhaps these could be captured as sensorimotor surfaces that are refined and established. Research on such processes could make use of the dynamical terminology introduced here, but would need to extend the terminology to capture how certain actions are reinforced

and coordinated patterns arise where they were not previously seen before.

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References

- Beer, R. D. (1995). A dynamical systems perspective on agent-environment interaction. *Artificial Intelligence*, 72(1), 173–215.
- Beer, R. D., & Gallagher, J. C. (1992). Evolving dynamical neural networks for adaptive behavior. *Adaptive Behavior*, 1(1), 91–122.
- Bryson, J. (2000). Hierarchy and sequence vs. full parallelism in action selection. *Proceedings of the Sixth International Conference on Simulation of Adaptive Behavior (SAB00)*, Paris, France, September 11–16.
- Buckley, C. L., Fine, P., Bullock, S., & Paolo, E. D. (2008). Monostable controllers for adaptive behaviour. In M. Asada, J. C. T. Hallam, J.-A. Meyer, & J. Tani (Eds.), *From animals to animats 10. Lecture Notes in Computer Science* (pp. 103–112). Berlin, Germany: Springer.
- Buhrmann, T., Di Paolo, E.A., & Barandiaran, X. (2013). A dynamical systems account of sensorimotor contingencies. *Frontiers in Psychology* 4.
- Clark, A. (1997). *Being there: Putting brain, body, and world together again*. Cambridge, MA: The MIT Press.
- Cliff, D., Husbands, P., & Harvey, I. (1993). Explorations in evolutionary robotics. *Adaptive Behavior*, 2(1), 73–110.
- Di Paolo, E. A., Noble, J., & Bullock, S. (2000). Simulation models as opaque thought experiments. *Artificial Life VII: The Seventh International Conference on the Simulation and Synthesis of Living Systems*, Reed College, Portland, Oregon, August 1–6.
- Egbert, M.D. (2013). Bacterial chemotaxis: Introverted or extroverted? A comparison of the advantages and disadvantages of basic forms of metabolism-based and metabolism-independent behavior using a computational model. *PloS One* 8(5), e63617.
- Harvey, I., Di Paolo, E., Wood, R., Quinn, M., Tuci, E., & Iridia, E. T. (2005). Evolutionary robotics: A new scientific tool for studying cognition. *Artificial Life*, 11(1–2), 79–98.
- Iizuka, H., & Di Paolo, E.A. (2007). Toward Spinozist robotics: Exploring the minimal dynamics of behavioral preference. *Adaptive Behavior*, 15(4): 359–376.
- Izquierdo, E., & Buhrmann, T. (2008). Analysis of a dynamical recurrent neural network evolved for two qualitatively different tasks: Walking and chemotaxis. In S. Bullock, J. Noble, R. Watson, & M. Bedau (Eds.), *Artificial Life XI: Proceedings of the Eleventh International Conference on the Simulation and Synthesis of Living Systems* (pp. 257–264). Cambridge, MA: The MIT Press.

- Izquierdo, E., Harvey, I., & Beer, R. D. (2008). Associative learning on a continuum in evolved dynamical neural networks. *Adaptive Behavior*, 16(6), 361–384.
- Kelso, J. A. (1995). *Dynamic patterns: The self organization of brain and behaviour*. Cambridge, MA: The MIT Press.
- Langton, C.G., ed. (1989). *Artificial life*. Redwood City, CA: Addison-Wesley.
- Lorenz, K. (1957). The nature of instinct. In C. H. Schiller (Ed.), *Instinctive behavior: The development of a modern concept* (pp. 129–175). New York: International Universities Press.
- Maes, P. (1991). A bottom-up mechanism for behavior selection in an artificial creature. *Proceedings of the First International Conference on Simulation of Adaptive Behavior on From Animals to Animats*. Cambridge, MA: The MIT Press.
- Nolfi, S., Floreano, D., Miglino, O., & Mondada, F. (1994, July). How to evolve autonomous robots: Different approaches in evolutionary robotics. In *Artificial Life IV* (pp. 190–197). Boston, MA.
- Port, R.F., & Van Gelder, T. eds. (1995). *Mind as motion: Explorations in the dynamics of cognition*. Cambridge, MA: The MIT Press.
- Schall, J. D. (2001). Neural basis of deciding, choosing and acting. *Nature Reviews Neuroscience*, 2(1), 33–42.
- Seth, A. K. (1998). Evolving action selection and selective attention without actions, attention, or selection. In *Proceedings of the Fifth International Conference on Simulation of Adaptive Behavior on From Animals to Animats (Vol. 5, pp. 139–146)*. Cambridge, MA: The MIT Press.
- Seth, A.K. (2007). The ecology of action selection: Insights from artificial life. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362(1485): 1545–1558.
- Slocum, A. C., Downey, D. C., & Beer, R. D. (2000, September). Further experiments in the evolution of minimally cognitive behavior: From perceiving affordances to selective attention. In *From Animals to Animats 6: Proceedings of the Sixth International Conference on Simulation of Adaptive Behavior* (pp. 430–439).
- Tinbergen, N. (1950). The hierarchical organization of nervous mechanisms underlying instinctive behaviour. *Symposia of the Society for Experimental Biology*, 4, 305–312.
- Tyrrell, T. (1993). *Computational mechanisms for action selection*. Doctoral dissertation, University of Edinburgh.
- Williams, P., & Beer, R.D. (2013). Environmental feedback drives multiple behaviors from the same neural circuit. In P. Lio, O. Miglino, G. Nicosia, S. Nolfi & M. Pavone (Eds.), *Advances in Artificial Life: ECAL 2013* (pp. 268–275). Cambridge, MA: The MIT Press.
- Williams, P., Beer, R., & Gasser, M. (2008, July). An embodied dynamical approach to relational categorization. In *Proceedings of the 30th Annual Conference of the Cognitive Science Society* (pp. 223–228), Nashville, Tennessee.
- Van Gelder, T. (1998). The dynamical hypothesis in cognitive science. *Behavioral and Brain Sciences*, 21(05), 615–628.
- Varela, F.J., Thompson, E.T., & Rosch, E. (1991). *The embodied mind: Cognitive science and human experience*. Cambridge, MA: The MIT Press.

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Randall D Beer received his PhD in computer science in 1989. From 1989 to 2006, he was a professor of electrical engineering and computer science, biology, and cognitive science at Case Western Reserve University. He spent the 1995–1996 academic year as a visiting scientist at the Santa Fe Institute, where he also served as an external faculty member for the next 6 years. In 2006, he joined the cognitive science program at Indiana University, where he is currently a professor of informatics and computing. Professor Beer's research is broadly concerned with understanding how coordinated behavior arises from the dynamical interaction of an animal's nervous system, its body, and its environment.

Appendix

The agents' environment consists of a bounded 100×100 unit plane, with two resources that are held in fixed locations throughout each trial. Each resource has a radius of 7 units. The concentration of chemical signals in the environment is determined by an exponential decay function of the distance from the center of the resource, $\text{concentration} = 10 \times e^{-0.05 \times \text{distance}}$. Within the spatial boundaries of a resource, the chemical concentration flattens out at a maximum value of 10. This introduces a sharp discontinuity in the dynamics, as

there is no unique derivative of the chemical concentration at the resources' boundaries. The chemical concentration is detected by the chemosensors, which extend outward from the agent's body by a fixed distance of 6 units, but the direction in which they project outward varies between the different morphologies.

The agent has two nutrient levels, which can take a continuous range of values between 0 and 10. A nutrient value is increased at an eating rate of 0.02 units per time step (integration time steps are set to 1 for all simulations) when the agent is within the boundary of

the nutrient's corresponding resource, and drained by 0.0045 units per time step through a constant metabolism regardless of position.

The outputs of the two effectors are used to generate movement through the environment. The pseudocode shown below describes the laws of motion for the agents. MaxAngle is equal to $\pi/12$ and MaxThrust is equal to 0.008. The friction coefficient of 0.9 makes it so the agent slows down and can eventually stop if the effectors produce no force.

$$\text{torque} = (\text{Output}(\text{Motor}_{\text{right}}) - \text{Output}(\text{Motor}_{\text{left}})) \times \text{MaxAngle}$$

$$\text{thrust} = (\text{Output}(\text{Motor}_{\text{right}}) + \text{Output}(\text{Motor}_{\text{left}})) \times \text{MaxThrust}$$

$$\text{velocity}_t = \text{velocity}_{t-1} \times 0.9 + \text{StepSize} \times \text{Thrust}$$

$$\text{angle}_t = \text{angle}_{t-1} + \text{StepSize} \times \text{Torque}$$

The nervous system, made of the sensors, effectors, and interneurons, is modeled by a CTRNN (Equation 1). In this equation, y is the state of each neuron, τ is the time constant, w_{ji} is the weight between neurons j and i , θ is a bias term, I is external input to

this neuron, and $\sigma(x) = 1/(1 + e^{-x})$ is the standard logistic activation function. For the chemosensors, I is set to the chemical concentration at their location. For the nutrient sensors I is set to the nutrient level. The rest of the CTRNN neurons have a constant I of 0.

$$\dot{y}_i = \frac{1}{\tau_i} \left(-y_i + \sum_{j=1}^N w_{ji} \sigma(y_j + \theta_j) + I_i \right) \quad (1)$$

A real-valued GA was applied to optimize longevity by searching through the CTRNN's parameter space. The evolutionary process begins with a population of random genetic strings, which encode the parameters of the CTRNN: τ , w , and θ . Each genetic string is a search vector of real numbers in the range ± 1 , and is scaled by each parameter's defined range. τ has the range of 1 to 10. w , and θ both have the range of -13 to 13 .

The agents' average longevity throughout the 11 trial configurations becomes their fitness value, and individuals are selected from the population with rank-based selection based on their fitness. The top-performing individual is copied twice into the next generation's population, and the rest of the population is repopulated through fitness-proportionate selection and mutation, with a mutation variance of 0.25.